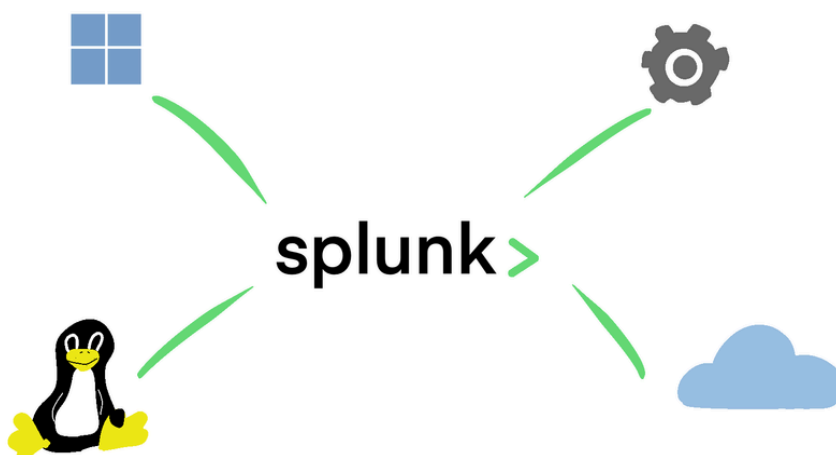


Build your own free SOC Lab with Splunk

 minder-security.ghost.io/get-started-with-splunk

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The term Splunk is associated with software solutions widely used in the fields of data analysis. In this tutorial, I will use what I learned during my time as a SOC Analyst to showcase how you can build your own SOC at home for free.

Lab Setup Tutorial

Our simplified Architecture will consist of the following components:

- **Ubuntu virtual machine** running Splunk Enterprise.
- **Windows 11 Enterprise (Evaluation) virtual machine** running Splunk Forwarder, which forwards data to our Ubuntu Splunk Enterprise Server.

1 - Installing Splunk Enterprise on Ubuntu

Scroll further down for a Cheat-Sheet of most useful Splunk CLI commands

After you setup your VM's, it's time to **Install Splunk Enterprise on our Ubuntu Virtual Machine**.

1. First, you need to create a free account to download the free trial version of Splunk Enterprise. I have decided to **download the .deb package**:

GET STARTED

Choose Your Download

Splunk Enterprise 9.2.1

Index 500 MB/Day. Sign up and download now. After 60 days you can convert to a perpetual free license or purchase a Splunk Enterprise license to continue using the expanded functionality designed for enterprise-scale deployments.

Choose Your Installation Package

Windows

Linux

Mac OS

64-bit

3.x+, 4.x+, or 5.4.x kernel
Linux distributions

.deb	520.37 MB	Download Now 	Copy wget link 	More ▾
.tgz	679.42 MB	Download Now 	Copy wget link 	More ▾
.rpm	679.24 MB	Download Now 	Copy wget link 	More ▾

[Release Notes](#) | [System Requirements](#) | [Previous Releases](#) | [All Other Downloads](#)

2. After the download was completed, I installed the package using the following command in the CLI:

Make sure **curl** is installed on your system!

```
sudo dpkg -i ./splunk<version>.deb
```

You can verify the installation using the following commands:

```
dpkg --status splunk  
dpkg --list
```

3. Then **start Splunk Enterprise for the first time** (note that the file path might vary depending on where you installed Splunk):

```
sudo /opt/splunk/bin/splunk start
```

4. You will be prompted to **accept the general terms**. Just press "q", then press "y" to accept the terms followed by pressing Enter:

SPLUNK GENERAL TERMS

Last Updated: August 12, 2021

These Splunk General Terms ("General Terms") between Splunk Inc., a Delaware corporation, with its principal place of business at 270 Brannan Street, San Francisco, California 94107, U.S.A ("Splunk" or "we" or "us" or "our") and you ("Customer" or "you" or "your") apply to the purchase of licenses and subscriptions for Splunk's Offerings. By clicking on the appropriate button, or by downloading, installing, accessing or using the Offerings, you agree to these General Terms. If you are entering into these General Terms on behalf of Customer, you represent that you have the authority to bind Customer. If you do not agree to these General Terms, or if you are not authorized to accept the General Terms on behalf of the Customer, do not download, install, access, or use any of the Offerings.

See the General Terms Definitions Exhibit attached for definitions of capitalized terms not defined herein.

1. License Rights

(A) General Rights. You have the nonexclusive, worldwide, nontransferable and nonsublicensable right, subject to payment of applicable Fees and compliance with the terms of these General Terms, to use your Purchased Offerings for Do you agree with this license? [y/n]: y

5. Then **setup the credentials** you want to use for your **Splunk Admin Account**. After you entered your credentials, Splunk Enterprise will be installed and you will be given a **Web-Interface URL**:

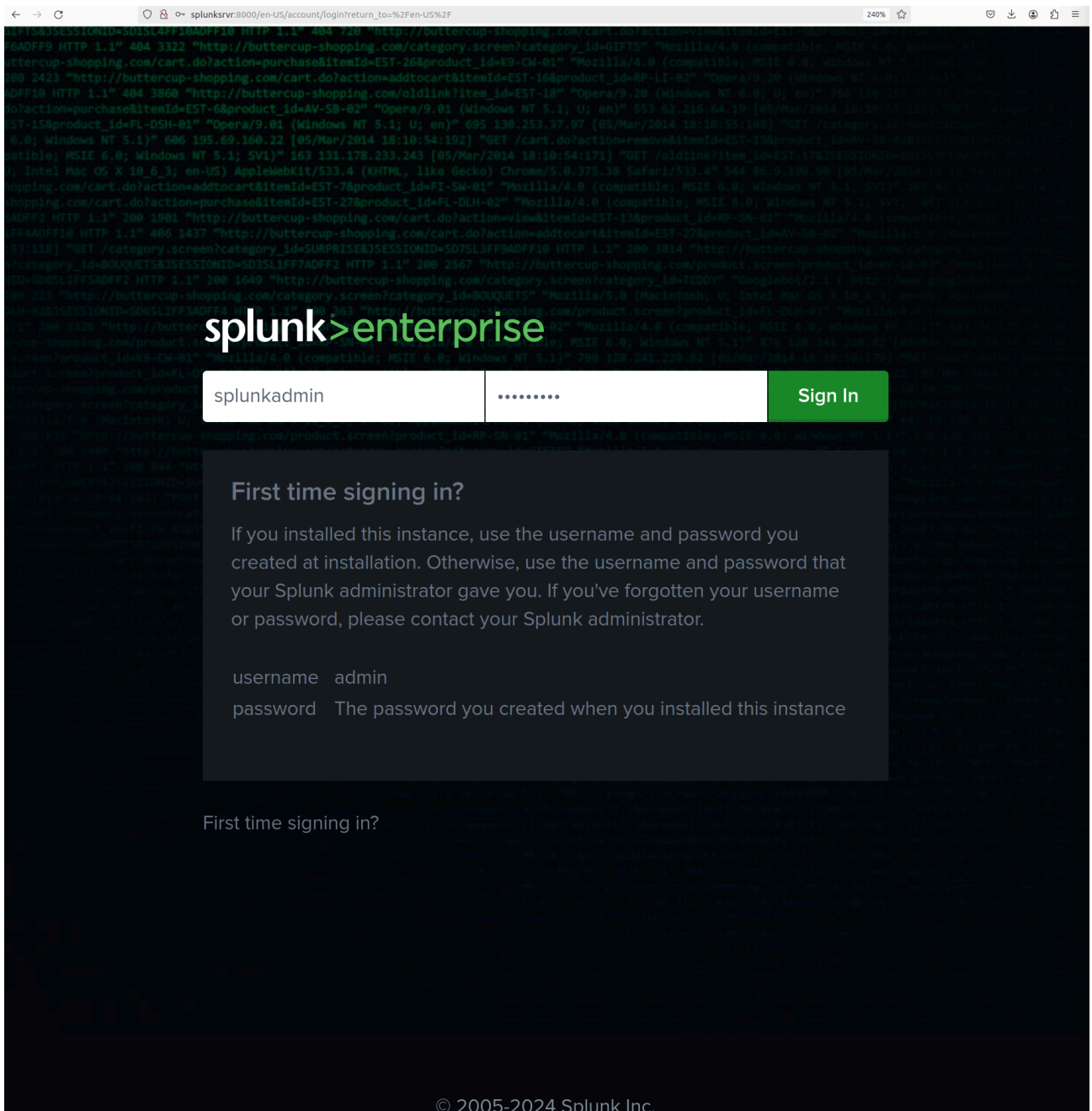
```
Starting splunk server daemon (splunkd)...
Generating a RSA private key
.....+++++
.....+++++
writing new private key to 'privKeySecure.pem'
-----
Signature ok
subject=/CN=splunksrvr/O=SplunkUser
Getting CA Private Key
writing RSA key
PYTHONHTTPSVERIFY is set to 0 in splunk-launch.conf disabling certificate validation for the httplib and urllib libraries shipped with the embedded Python interpreter; must be set to "1" for increased security
Done

Waiting for web server at http://127.0.0.1:8000 to be available..... Done

If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://splunksrvr:8000
```

6. **Open the URL inside your Webbrowser**. You will be greeted by a login-page looking similar to this one. Use your credentials so authenticate:



7. After authenticating, you will be greeted by the **Splunk Enterprise Homepage**:

splunk>enterprise

Apps

Administrator

Messages

Settings

Activity

Help

Find

Apps

Manage

Search apps by name...

Search & Reporting

Splunk Secure Gateway

Upgrade Readiness App

Find more apps

Hello, Administrator

Quick links

Dashboard

Recently viewed

Created by you

Shared with you

Common tasks

Add data

Add data from a variety of common sources.

Search your data

Turn data into doing with Splunk search.

Visualize your data

Create dashboards that work for your data.

Add team members

Add your team members to Splunk platform.

Manage permissions

Control who has access with roles.

Configure mobile devices

Login or manage mobile devices using Splunk Secure Gateway.

Learning and resources

Product tours

New to Splunk? Take a tour to help you on your way.

Learn more with Splunk Docs

Deploy, manage, and use Splunk software with comprehensive guidance.

Get help from Splunk experts

Actionable guidance on the Splunk Lantern Customer Success Center.

Extend your capabilities

Browse thousands of apps on Splunkbase.

Join the Splunk Community

Learn, get inspired, and share knowledge.

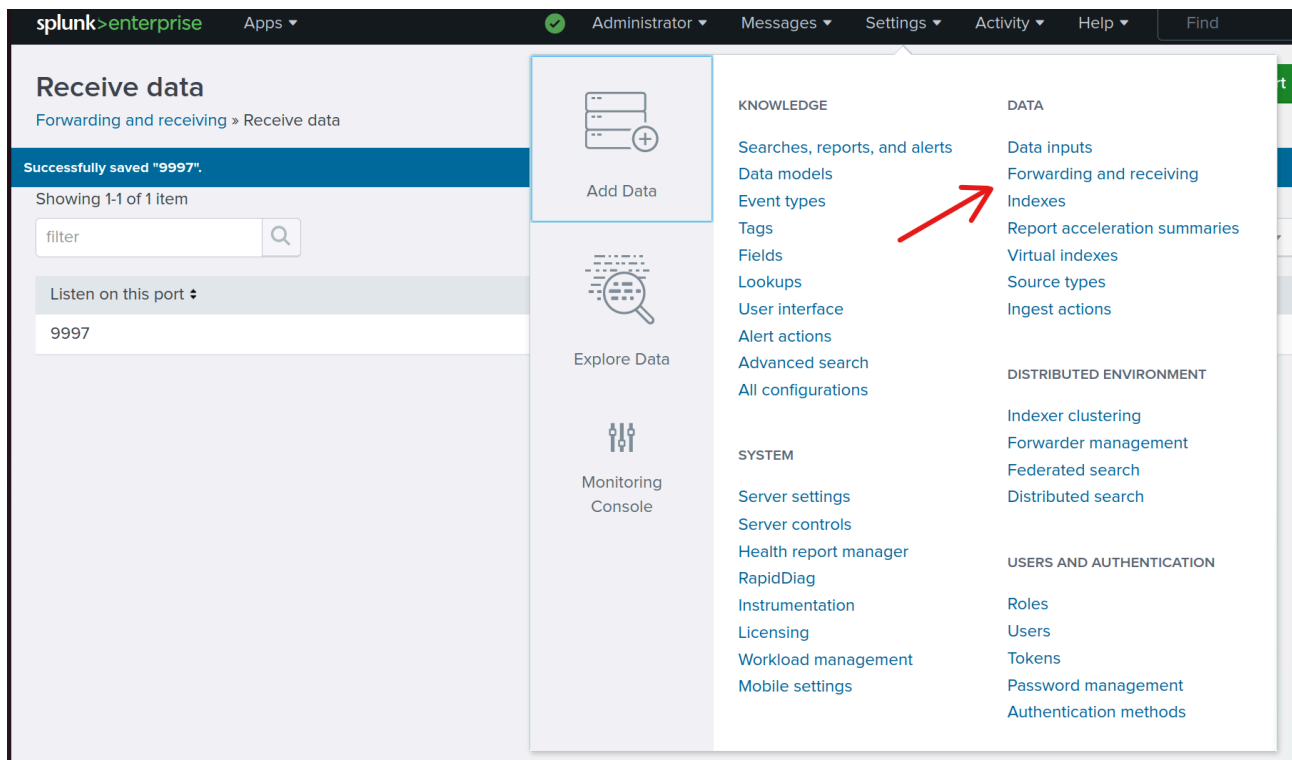
See how others use Splunk

Browse real customer stories.

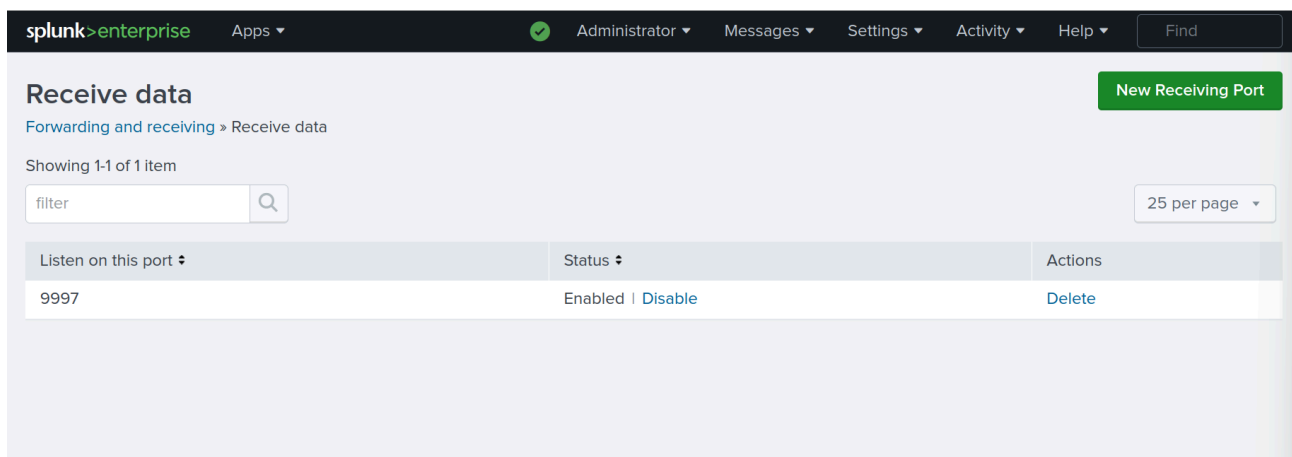
Training and Certification

Become a certified Splunk Ninja.

8. **Configure Splunk Enterprise to receive Data from Forwarders** by going into Settings>Forwarding and receiving:



- Click on **Configure receiving > New Receiving Port**. Enter the port you wish to use to receive Data from Forwarders. I chose the default Port 9997. Click on **Save**, then you should see the following configuration on your page:



- Verify that your Listening Port is enabled, then check the IP-Address of your Splunk Server using the following command:

```
ip addr
```

```
xhor@splunksrvr:~/Downloads$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:bb:29:e7 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.137.148/24 brd 192.168.137.255 scope global dynamic noprefixroute ens33
        valid_lft 601055sec preferred_lft 601055sec
    inet6 fe80::4065:86b2:1269:9fdb/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Note down the IP-Address your server uses.

I'm using a Bridged Network Setup for my Ubuntu VM

2 - Forwarding Data from Windows Hosts

After Splunk Enterprise has been installed on our Ubuntu Server, it's time to forward data to it. We will do so using **Splunk Universal Forwarder**. Follow these steps:

1. **Download the Splunk Universal Forwarder onto the Windows Host.** Again, do so using your free account. I installed the 64-Bit Version:

Splunk Universal Forwarder 9.2.1

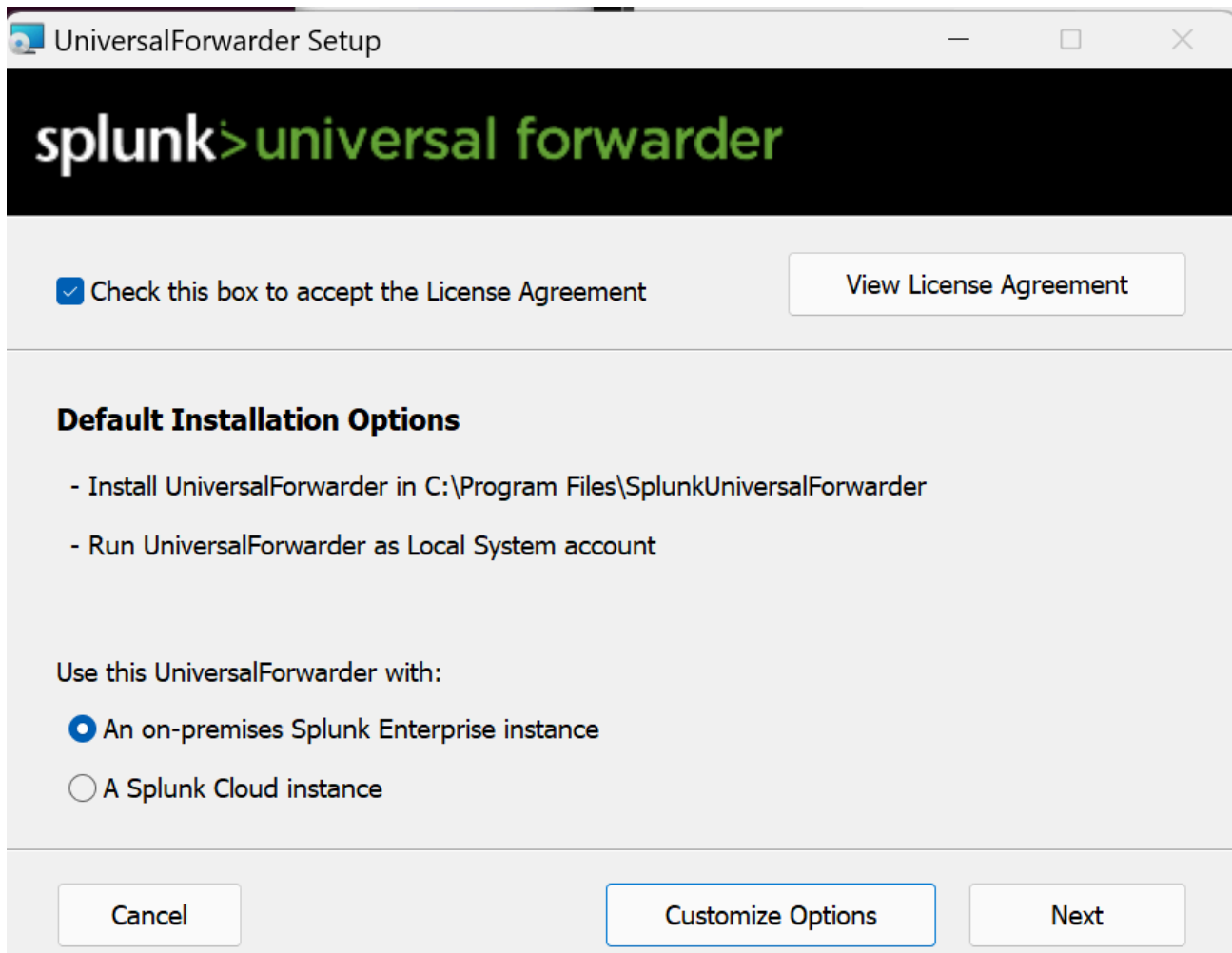
Universal Forwarders provide reliable, secure data collection from remote sources and forward that data into Splunk software for indexing and consolidation. They can scale to tens of thousands of remote systems, collecting terabytes of data.

Choose Your Installation Package

OS	Architecture	Package Name	Size	Download Now	Copy wget link	More	
Windows	32-bit	Windows 10	.msi	64.29 MB	Download Now	Copy wget link	More
Windows	64-bit	Windows 10, Windows 11 Windows Server 2016, 2019, 2022	.msi	118.96 MB	Download Now	Copy wget link	More

[Release Notes](#) | [System Requirements](#) | [Previous Releases](#) | [All Other Downloads](#)

2. **Execute the downloaded .msi File** on your Windows host. Check the License Agreement Box. Then click **Customize Options**:



3. Click **Next** until you get to the page shown in the picture below. **Select all the data sources you want to collect.** I chose everything, just for the sake of curiosity. Then click on **Next**:

UniversalForwarder Setup

splunk>universal forwarder

Windows Event Logs	Performance Monitor
☒ Application Logs	☒ CPU Load
☒ Security Log	☒ Memory
☒ System Log	☒ Disk Space
☒ Forwarded Events Log	☒ Network Stats
☒ Setup Log	

Active Directory Monitoring

☐ Enable AD monitoring

Path to monitor

4. **Create an administrator account for the forwarder.** Click **Next** after you're done:

UniversalForwarder Setup

splunk>universal forwarder

Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.

Username:

SplunkForwarderAdmin

☐ Generate random password

Password:

Confirm password:

Cancel Back Next

5. In this step, we can configure a deployment server, which is useful when you have multiple hosts to manage. Enter the **hostname/IP of your Splunk Enterprise Server** and the **Port** (default if you haven't configured anything else). Then proceed by **clicking Next**.

UniversalForwarder Setup

splunk>universal forwarder

If you intend to use a Splunk deployment server to configure this UniversalForwarder, please specify the host or IP, and port (default port is 8089). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.

Deployment Server

Hostname or IP

:

Enter the hostname or IP of your deployment server, e.g. ds.splunk.com *default is 8089*

6. Now enter the hostname/IP of your Splunk Enterprise Server, as well as the Port you configured for receiving Data:

UniversalForwarder Setup

splunk>universal forwarder

If you intend to use a Splunk receiving indexer to configure this UniversalForwarder, please specify the host or IP, and port (default port is 9997). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.

Receiving Indexer

Hostname or IP

192.168.137.148 : 9997

Enter the hostname or IP of your receiving indexer, e.g. ds.splunk.com *default is 9997*

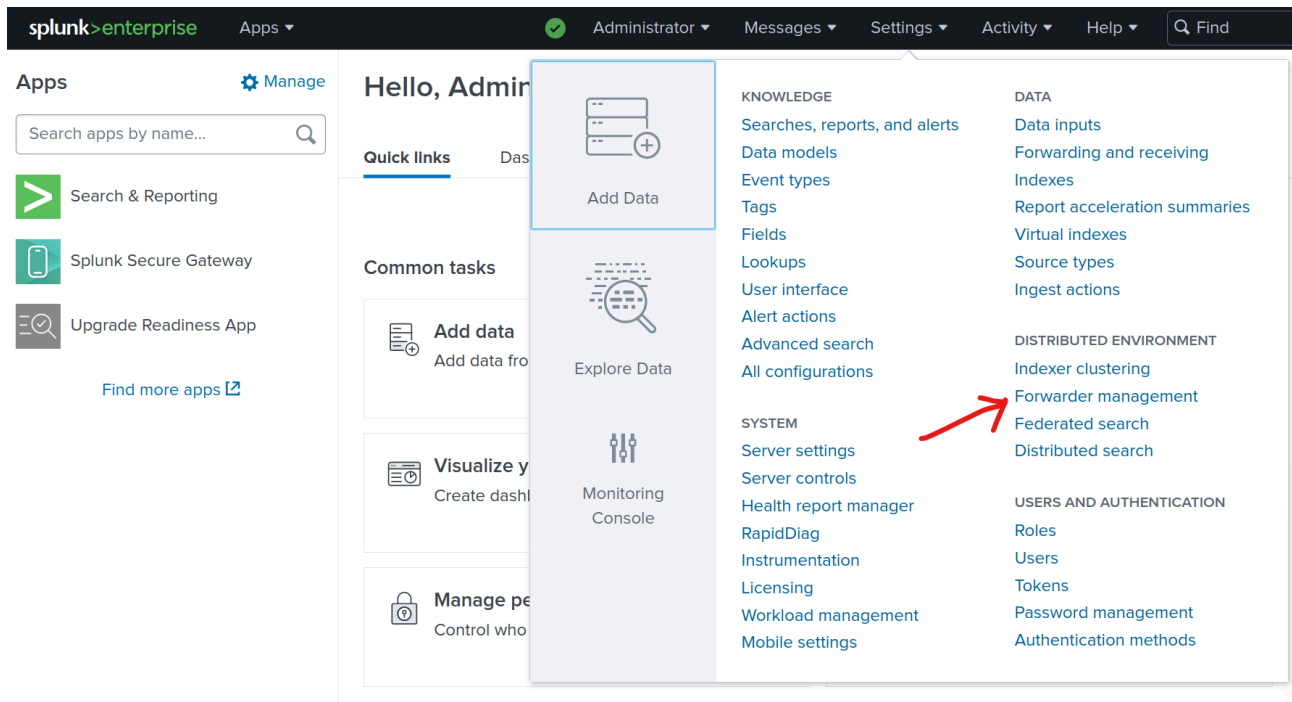
Cancel Back Next

7. Click on **Install**, then **Finish**.

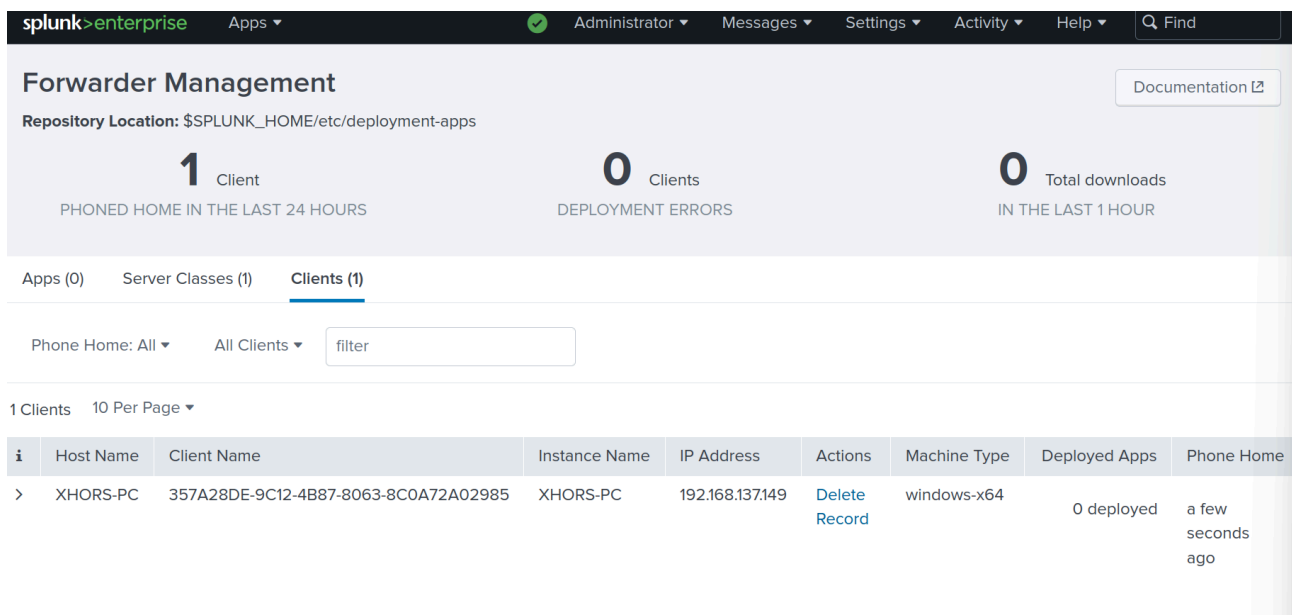
To uninstall your configuration, stop the service from running and uninstall Splunk Universal Forwarder in your Windows App-Settings.

3 - Testing your Setup

Now you have successfully configured the client to forward Data to our Splunk Enterprise Server running on Ubuntu Linux. If you configured the **Deployment Server**, you can verify that the Forwarder has connected to the Server by opening **Settings > Forwarder Management**:



Then you should be able to see your onboarded Device:



To run Splunk Processing Language (SPL) Queries, you can go to **Apps > Search & Reporting** and start querying data:

The screenshot shows the Splunk Enterprise App Manager interface. The top navigation bar includes 'splunk>enterprise', 'Apps', 'Administrator', 'Messages', 'Settings', 'Activity', 'Help', and a search bar. A dropdown menu is open under 'Apps', showing options: 'Search & Reporting', 'Splunk Secure Gateway', 'Upgrade Readiness App', 'Manage Apps', and 'Find More Apps'. The main content area shows '0 Clients' and '0 Total downloads IN THE LAST 1 HOUR'. Below this, there are tabs for 'Apps (0)', 'Server Classes (1)', and 'Clients (1)'. The 'Clients (1)' tab is active, showing a table with one client: 'XHORS-PC'. The table columns are: Host Name, Client Name, Instance Name, IP Address, Actions, Machine Type, Deployed Apps, and Phone Home. The 'Actions' column for the client shows 'Delete' and 'Record' links.

You can start by typing the following line into the search bar:

```
index="main" source="wineventlog:security"
```

This line will return all logs collected from our Windows Security Event Logs. You can start to play around with the GUI and try out some SPL queries:

splunk>enterprise Apps Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Search & Reporting

New Search

Save As Create Table View Close

index="main" source="wineventlog:security" Last 24 hours

✓ 2,857 events (4/5/24 1:00:00.000 PM to 4/6/24 1:09:37.000 PM) No Event Sampling Job

Events (2,857) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

List Format 20 Per Page < Prev 1 2 3 4 5 6 7 8 ... Next >

< Hide Fields All Fields

SELECTED FIELDS

- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- a Account_Domain 5
- a Account_Name 11
- a ComputerName 1
- # EventCode 20
- # EventType 1
- a index 1
- a Keywords 1
- # linecount 15
- a LogName 1
- a Logon_ID 10
- a Message 54
- a OpCode 1
- a punct 21
- a Read_Operation 2
- # RecordNumber 100+
- a Security_ID 10
- a SourceName 1
- a splunk_server 1
- a TaskCategory 10
- a Type 1

62 more fields

+ Extract New Fields

i	Time	Event
>	4/6/24 12:53:08.000 PM	04/06/2024 12:53:08 PM LogName=Security EventCode=4672 EventType=0 ComputerName=XHORS-PC Show all 31 lines host = XHORS-PC source = WinEventLog:Security sourcetype = WinEventLog:Security
>	4/6/24 12:53:08.000 PM	04/06/2024 12:53:08 PM LogName=Security EventCode=4624 EventType=0 ComputerName=XHORS-PC Show all 71 lines host = XHORS-PC source = WinEventLog:Security sourcetype = WinEventLog:Security
>	4/6/24 12:45:38.000 PM	04/06/2024 12:45:38 PM LogName=Security EventCode=4672 EventType=0 ComputerName=XHORS-PC Show all 22 lines host = XHORS-PC source = WinEventLog:Security sourcetype = WinEventLog:Security
>	4/6/24 12:45:38.000 PM	04/06/2024 12:45:38 PM LogName=Security EventCode=4624 EventType=0 ComputerName=XHORS-PC Show all 71 lines host = XHORS-PC source = WinEventLog:Security sourcetype = WinEventLog:Security

Now you have everything you need to get started with Splunk Enterprise!

Thank you for reading my blog post and I hope I could help you build your own SOC Lab!

Additional Info: Splunk CLI Cheat Sheet

Starting, Stopping, Restarting and checking Status of the Splunk Enterprise Server

```
splunk start
```

```
splunk stop
```

```
splunk restart
```

```
splunk status
```

Add a Test-Event to the Splunk Index

```
splunk add oneshot
```

Search Data in the Splunk Index

```
splunk search <keyword>
```

More Information

```
splunk help
```